Thomas Diederen

Computational biologist

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Summary

Computational biologist at the interface of machine learning and biological modelling. Experience in simulation-based inference, combining mechanistic models with modern generative ML approaches. Worked extensively with omics data (metabolomics, proteomics, genomics) in modelling and inference pipelines. Interested in developing machine learning methods for complex biological systems in drug discovery.

Experience

Consultant, computational biology — Hybrid, Munich, Bavaria
Differential Bio — Jul. 2025 - Mar. 2026 (contract)

- Built a fully differentiable dynamic flux balance analysis (dFBA) simulator in JAX enabling Bayesian inference over 2,100 batch fermentations from OD600 data alone.
- Designed an automated genome-scale metabolic model reconstruction and curation pipeline; contributed to the company's first commercial software license.
- Translated client fermentation objectives into modelling requirements, scoped deliverables, and drove adoption of model-based medium design tools across experimental teams.

Personal goal pursuit — Mediterranean coast
Traveling & climbing — Feb. 2024 – Dec. 2024

Ph.D. titled: Simulation-based Metabolic Flux Inference — Zürich, Switzerland
Zamboni lab, Institute for Molecular Systems Biology, ETH Zürich — Aug. 2018 – Jan. 2024

- Pioneered methods for modeling Riemannian continuous normalizing flows on convex polytopes (paper link).
- Developed Bayesian inference pipelines for biochemical reaction rates from metabolomics data (LC-MS).
- Integrated metabolomics data with genome-scale models for metabolic flux analysis.
- Engineered the `sbmfi` package for simulation-based metabolic flux inference (PyTorch).

Skills

Modelling	Normalizing flows, diffusion models, variational inference, Bayesian neural networks, MCMC, convex optimization, ODE-based modelling, statistical modelling of omics data
Programming	Python (advanced), Matlab, JavaScript, R, C++, Bash
Packages	PyTorch, JAX, NumPy, SciPy, scikit-learn, Pandas, Matplotlib, COBRAPy
Data	Metabolomics (LC-MS), genomics, proteomics, biochemical assays
General	Git, Docker, LaTeX, claude code, cursor, MLFlow, Ray, Hetzner deployment
Languages	Dutch (fluent), Polish (fluent), English (fluent), German (B2)

Education

M.Sc. Biotechnology - Systems Biology — Basel, Switzerland
Department of Biosystems Science and Engineering, ETH Zürich — Sept. 2015 - Aug. 2017

- Thesis: Insulation in the Stochastic Setting, Control Theory and Systems Biology Laboratory

B.Sc. Life Science & Technology — Delft, The Netherlands
Delft University of Technology — Sept. 2011 - Nov. 2014